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FUEL-AIR MIXER FOR TURBINE ENGINE COMBUSTION SECTION

Abstract

An apparatus is provided for a turbine engine. This apparatus includes a fuel-air mixer, and the fuel-air mixer includes a mixer outlet, an annular inner passage, an air swirler, an annular fuel passage and a fuel swirler. The annular inner passage extends axially along an axis within the fuel-air mixer. The annular inner passage includes an inner passage downstream section and an inner passage upstream section fluidly coupled with the mixer outlet through the inner passage downstream section. The inner passage downstream section radially tapers and diverges radially outward away from the axis as the annular inner passage extends axially towards the mixer outlet. The air swirler is disposed with the inner passage upstream section. The annular fuel passage circumscribes the annular inner passage and extends axially within the fuel-air mixer to the inner passage downstream section. The fuel swirler is disposed with the annular fuel passage.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

1. Technical Field

[0002] This disclosure relates generally to a turbine engine and, more particularly, to a fuel-air mixer for the turbine engine.

2. Background Information

[0003] As government emissions standards tighten, interest in alternative fuels for gas turbine engines continues to grow. There is interest, for example, in fueling a gas turbine engine with hydrogen (H.sub.2) fuel rather than a traditional hydrocarbon fuel such as kerosine to reduce greenhouse emissions. While known hydrogen combustion systems have various advantages, there is still room in the art for improvement.

SUMMARY OF THE DISCLOSURE

[0004] According to an aspect of the present disclosure, an apparatus is provided for a turbine engine. This apparatus includes a fuel-air mixer, and the fuel-air mixer includes a mixer outlet, an annular inner passage, an air swirler, an annular fuel passage and a fuel swirler. The annular inner passage extends axially along an axis within the fuel-air mixer. The annular inner passage includes an inner passage downstream section and an inner passage upstream section fluidly coupled with the mixer outlet through the inner passage downstream section. The inner passage downstream section radially tapers and diverges radially outward away from the axis as the annular inner passage extends axially towards the mixer outlet. The air swirler is disposed with the inner passage upstream section. The annular fuel passage circumscribes the annular inner passage and extends axially within the fuel-air mixer to the inner passage downstream section. The fuel swirler is disposed with the annular fuel passage.

[0005] According to another aspect of the present disclosure, another apparatus is provided for a turbine engine. This apparatus includes a fuel-air mixer, and the fuel-air mixer includes an annular first passage, an annular inner passage, an annular outer passage and an annular fuel passage. The annular inner passage extends axially along an axis within the fuel-air mixer to the annular first passage. The annular inner passage diverges radially outward away from the axis as the annular inner passage extends axially to the annular first passage. The annular outer passage extends axially along the axis within the fuel-air mixer to the annular first passage. The annular outer passage circumscribes the annular inner passage and the annular fuel passage. The annular outer passage converges radially inward towards the axis as the annular outer passage extends axially to the annular first passage. The annular fuel passage is radially between the annular inner passage and the annular outer passage. The annular fuel passage circumscribes the annular inner passage. The annular fuel passage extends axially along the axis within the fuel-air mixer to an annular outlet from the annular fuel passage into the annular inner passage.

[0006] According to still another aspect of the present disclosure, another apparatus is provided for a turbine engine. This apparatus includes a fuel-air mixer, and the fuel-air mixer includes a mixer outlet, an annular inner passage, an annular fuel passage and a resonator. The annular inner passage extends axially along an axis within the fuel-air mixer and circumscribes the resonator. The annular inner passage includes an inner passage downstream section and an inner passage upstream section fluidly coupled with the mixer outlet through the inner passage downstream section. The inner passage downstream section diverges radially outward away from the axis as the annular inner passage extends axially towards the mixer outlet. The annular fuel passage circumscribes the annular inner passage and extends axially within the fuel-air mixer to the inner passage downstream section. The resonator forms an inner peripheral boundary of the annular inner passage. The resonator includes an air inlet, a plurality of air outlets and an internal cavity fluidly

coupled with and between the air inlet and the air outlets. The internal cavity fluidly coupled with the mixer outlet through the air outlets.

[0007] A cross-sectional area of the inner passage downstream section may decrease as the annular inner passage extends axially towards the mixer outlet.

[0008] A centerline of a half of the inner passage upstream section may be parallel with the axis. A centerline of a half of the inner passage downstream section may be angularly offset from the axis by an acute angle.

[0009] A centerline of a half of the annular fuel passage may be angularly offset from a centerline of a half of the inner passage downstream section by an acute angle.

[0010] A centerline of a half of the annular fuel passage at an annular outlet from the annular fuel passage into the inner passage downstream section may be parallel with the axis.

[0011] The air swirler may be configured to swirl air flowing within the annular inner passage towards the mixer outlet in a first circumferential direction about the axis. The fuel swirler may be configured to swirl fuel flowing within the annular fuel passage towards the inner passage downstream section in the first circumferential direction about the axis.

[0012] The fuel-air mixer may also include an annular outer passage circumscribing the annular inner passage and the annular fuel passage. The annular outer passage may extend axially along the axis within the fuel-air mixer and is fluidly coupled with the mixer outlet.

[0013] The annular outer passage may converge radially inward towards the axis as the annular outer passage extends axially towards the mixer outlet.

[0014] The fuel-air mixer may also include a plurality of non-swirling struts arranged circumferentially about the axis and extending radially across the annular outer passage.

[0015] The air swirler may be an inner air swirler. The fuel-air mixer may also include an outer air swirler disposed with the annular outer passage.

[0016] The fuel-air mixer may also include a center body with the annular inner passage extending axially along and circumscribing the center body. The center body may include an air inlet, a plurality of air outlets and an internal cavity fluidly coupled with and between the air inlet and the air outlets. The internal cavity may be fluidly coupled with the mixer outlet through the air outlets.

[0017] The center body may form an inner peripheral boundary of the annular inner passage.

[0018] A cross-sectional area of the air inlet may be greater than a cross-sectional area of each of the air outlets. The cross-sectional area of the air inlet may be less than a total cross-sectional area of the air outlets.

[0019] The internal cavity may include a cavity upstream section and a cavity downstream section between the cavity upstream section and the air outlets. The cavity downstream section may radially expand as the internal cavity extends axially within the center body towards the air outlets.

[0020] The apparatus may also include an annular combustor bulkhead extending circumferentially around an axial centerline. The fuel-air mixer may be one of a plurality of fuel-air mixers mounted to the annular combustor bulkhead. A first of the fuel-air mixers may be located radially outboard of a second of the fuel-air mixers.

[0021] The first of the fuel-air mixers may be circumferentially aligned with the second of the fuel-air mixers.

[0022] The apparatus may also include a pilot fuel injector. The fuel-air mixer may be one of a plurality of fuel-air mixers arranged in an array symmetrically about the pilot fuel injector.

[0023] The apparatus may also include a pilot fuel injector. The fuel-air mixer may be one of a plurality of fuel-air mixers arranged in an array asymmetrically about the pilot fuel injector.

[0024] The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.

[0025] The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. 1 is a partial schematic illustration of a gas turbine engine.

[0027] FIG. 2 is a partial schematic illustration of a combustor section between a compressor section and a turbine section.

[0028] FIG. 3 is a schematic end view illustration of a combustor with multiple fuel-air mixers.

[0029] FIG. 4 is a partial sectional illustration of a combustor wall.

[0030] FIG. 5 is a partial schematic illustration of another combustor with the fuel-air mixers.

[0031] FIG. 6 is a partial sectional illustration of one of the fuel-air mixers.

[0032] FIGS. 8A and 8B are partial schematic illustrations of the fuel-air mixers in various arrangements with pilot fuel injectors.

[0033] FIG. 7 is an end view illustration of one of the fuel-air mixers.

[0034] FIG. 9 is a partial sectional illustration of another fuel-air mixer.

DETAILED DESCRIPTION

[0035] FIG. 1 illustrates a gas turbine engine 20 for an aircraft. The aircraft may be an airplane, a drone (e.g., an unmanned aerial vehicle (UAV)) or any other manned or unmanned aerial vehicle or system. The turbine engine 20 of FIG. 1 is configured as, or may be part of, a propulsion system for the aircraft. The turbine engine 20, however, may also or alternatively be configured as, or may be part of, an electrical power system for the aircraft. The turbine engine 20 of the present disclosure, however, is not limited to aircraft applications. The turbine engine 20, for example, may alternatively be configured as an industrial turbine engine.

[0036] The turbine engine 20 of FIG. 1 extends axially along an axial centerline 22 between a forward, upstream end 24 of the turbine engine 20 and an aft, downstream end 26 of the turbine engine 20. The turbine engine 20 includes a fan section 28, a compressor section 29, a combustor section 30 and a turbine section 31. The compressor section 29 of FIG. 1 includes a low pressure compressor (LPC) section 29A and a high pressure compressor (HPC) section 29B. The turbine section 31 of FIG. 1 includes a high pressure turbine (HPT) section 31A and a low pressure turbine (LPT) section 31B.

[0037] The engine sections 28-31B of FIG. 1 are arranged sequentially along the axial centerline 22 within an engine housing 32. This engine housing 32 includes an inner case 34 (e.g., a core case) and an outer case 36 (e.g., a fan case). The inner case 34 may house one or more of the engine sections 29A-31B; e.g., a core of the turbine engine 20. The outer case 36 may house at least the fan section 28.

[0038] Each of the engine sections 28, 29A, 29B, 31A and 31B includes a respective bladed rotor 38-42. Each of these bladed rotors 38-42 includes a plurality of rotor blades arranged circumferentially around and connected to one or more respective rotor disks and/or hubs. The rotor blades, for example, may be formed integral with or mechanically fastened, welded, brazed and/or otherwise attached to the respective rotor disk(s) and/or the respective hub(s).

[0039] The fan rotor 38 is connected to a geartrain 44, for example, through a fan shaft 46. The geartrain 44 and the LPC rotor 39 are connected to and driven by the LPT rotor 42 through a low speed shaft 47. The HPC rotor 40 is connected to and driven by the HPT rotor 41 through a high speed shaft 48. The engine shafts 46-48 are rotatably supported by a plurality of bearings; e.g., rolling element and/or thrust bearings. Each of these bearings is connected to the engine housing 32 by at least one stationary structure such as, for example, an annular support strut.

[0040] During engine operation, air enters the turbine engine 20 through an airflow inlet 50 into the turbine engine 20. This air is directed through the fan section 28 and into a (e.g., annular) core flowpath 52 and a (e.g., annular) bypass flowpath 54. The core flowpath 52 extends sequentially through the engine sections 29A-31B (e.g., the engine core) from an inlet 56 into the core flowpath

52 to an exhaust **58** from the core flowpath **52**. The air within the core flowpath **52** may be referred to as “core air”. The bypass flowpath **54** extends through a bypass duct, and bypasses (e.g., extends axially along and outside of) the engine core. The air within the bypass flowpath **54** may be referred to as “bypass air”.

[0041] The core air is compressed by the LPC rotor **39** and the HPC rotor **40** and directed into a (e.g., annular) combustion chamber **60** within the combustor section **30**. Fuel is injected into the combustion chamber **60** and mixed with the compressed core air to provide a fuel-air mixture. This fuel-air mixture is ignited and combustion products thereof flow through and sequentially cause the HPT rotor **41** and the LPT rotor **42** to rotate before being directed out of the turbine engine **20** through the core exhaust **58**. The rotation of the HPT rotor **41** and the LPT rotor **42** respectively drive rotation of the HPC rotor **40** and the LPC rotor **39** and, thus, compression of the air received from the core inlet **56**. The rotation of the LPT rotor **42** also drives rotation of the fan rotor **38**, which propels the bypass air through the bypass flowpath **54** and out of the turbine engine **20** through an exhaust **65** from the bypass flowpath **54**. The propulsion of the bypass air may account for a majority of thrust generated by the turbine engine **20**.

[0042] FIG. **2** illustrates a portion of the combustor section **30** along the core flowpath **52** between the HPC section **29B** and the HPT section **31A**. This combustor section **30** includes a diffuser plenum **62**, a combustor **64** and one or more fuel-air mixers **66A** and **66B** (generally referred to as “**66**”); see also FIG. **3**. Briefly, the combustor **64** and the fuel-air mixers **66** are disposed within (e.g., surrounded by) the diffuser plenum **62**. The diffuser plenum **62** is configured to receive compressed core air from the HPC section **29B** for subsequent provision into the combustion chamber **60**, for example, substantially through the fuel-air mixers **66**.

[0043] The combustor **64** may be configured as an annular combustor; e.g., an annular floating wall combustor. The combustor **64** of FIGS. **2** and **3**, for example, includes an annular combustor bulkhead wall **68** (“bulkhead”), a tubular inner combustor wall **70A** (“inner wall”) and a tubular outer combustor wall **70B** (“outer wall”). The bulkhead **68** of FIG. **2** extends radially between and to the inner wall **70A** and the outer wall **70B**. The bulkhead **68** may be connected (e.g., mechanically fastened or otherwise attached) to the inner wall **70A** and/or the outer wall **70B**. Each combustor wall **70A**, **70B** (generally referred to as “**70**”) projects axially along the axial centerline **22** out from the bulkhead **68** towards the HPT section **31A**. The inner wall **70A** of FIG. **2**, for example, projects axially to and may be connected to an inner platform **72A** of a downstream stator vane array **74**; e.g., a turbine inlet nozzle in the HPT section **31A**. The outer wall **70B** of FIG. **2** projects axially to and may be connected to an outer platform **72B** of the stator vane array **74**. With the arrangement of FIG. **2**, the combustion chamber **60** is formed by and extends radially within the combustor **64** between and to the inner wall **70A** and the outer wall **70B**. The combustion chamber **60** is formed by and extends axially (in an upstream direction along the core flowpath **52**) into the combustor **64** from the stator vane array **74** to the bulkhead **68**. The combustion chamber **60** also extends within the combustor **64** circumferentially about (e.g., completely around) the axial centerline **22**.

[0044] Referring to FIG. **4**, any one or more or all of the combustor walls **68**, **70A**, **70B** may each be configured as a multi-walled structure; e.g., a hollow, dual-walled structure. For example, each combustor walls **68**, **70A**, **70B** of FIG. **4** includes a combustor wall shell **76**, a combustor wall heat shield **78** (e.g., a liner) and one or more combustor wall cooling cavities **80** (e.g., impingement cavities) formed by and (e.g., radially and/or axially) between the shell **76** and the heat shield **78**. Each cooling cavity **80** of FIG. **4** is fluidly coupled with the diffuser plenum **62** through one or more cooling apertures **82** in the shell **76**; e.g., impingement apertures. Each cooling cavity **80** of FIG. **4** is fluidly coupled with the combustion chamber **60** through one or more cooling apertures **84** in the heat shield **78**; e.g., effusion apertures. Of course, various other multi-walled combustor wall structures are known in the art, and the present disclosure is not limited to any particular ones thereof. Furthermore, it is contemplated any one or more of the combustor walls **68**, **70A**, **70B** of

FIG. 2 may alternatively be configured as a single-walled structure. The shell **76** (see FIG. 4) for example, may be omitted and the heat shield **78** may form a single walled liner/wall. Referring to FIG. 5, it is also contemplated any one or more of the combustor walls **68**, **70A**, **70B** may alternatively be configured as a hybrid between a multi-walled structure and a single walled structure. Each combustor wall **70A**, **70B** of FIG. 5, for example, includes an upstream section **86A**, **86B** and a downstream section **88A**, **88B**, where the upstream section **86A**, **86B** is configured as a multi-walled structure and the downstream section **88A**, **88B** is configured as a single walled structure.

[0045] Referring to FIG. 2, the stator vane array **74** includes the inner platform **72A**, the outer platform **72B** and a plurality of stator vanes **90** (one visible in FIG. 2). The stator vanes **90** are arranged circumferentially about the axial centerline **22** in an array; e.g., a circular array. Each of these stator vanes **90** extends radially across the core flowpath **52** between and to the inner platform **72A** and the outer platform **72B**. Each of the stator vanes **90** may also be connected to the inner platform **72A** and/or the outer platform **72B**. The stator vane array **74** and its stator vanes **90** are configured to turn and/or otherwise condition the combustion products exiting the combustion chamber **60** for interaction with a first stage of the HPT rotor **41**; see FIG. 1.

[0046] Referring to FIG. 3, the fuel-air mixers **66** may be divided into one or more sets of the fuel-air mixers **66**. The inner fuel-air mixers **66A** in the inner set are arranged circumferentially about the axial centerline **22** in an inner mixer array; e.g., a circular array. The outer fuel-air mixers **66B** in the outer set are arranged circumferentially about the axial centerline **22** in an outer mixer array; e.g., a circular array. Within each array, the fuel-air mixers **66** may be equally spaced about the axial centerline **22** by a common (e.g., the same) circumferential inter-mixer distance. Each outer fuel-air mixer **66B** is circumferentially aligned with a respective one of the inner fuel-air mixers **66A**. Each inner fuel-air mixer **66A** of FIG. 3 is similarly circumferentially aligned with a respective one of the outer fuel-air mixers **66B**. The present disclosure, however, is not limited to such an exemplary fuel-air mixer arrangement. For example, one or more of the outer fuel-air mixers **66B** may be circumferentially offset from the circumferentially closest inner fuel-air mixer **66A** and/or one or more of the inner fuel-air mixers **66A** may be circumferentially offset from the circumferentially closest outer fuel-air mixer **66B**. In another example, the fuel-air mixers **66** within each array may be arranged into a plurality of groupings; e.g., pairs. The groupings may be spaced circumferentially about the axial centerline **22** by a common first circumferential inter-grouping distance. The fuel-air mixers **66** within each grouping may be spaced about the axial centerline **22** by a common circumferential intra-grouping distance that is different (e.g., less) than the inter-grouping distance.

[0047] Referring to FIG. 2, each fuel-air mixer **66** includes a fuel conduit **92** and a tubular mixer body **94**. The fuel conduit **92** is configured to route the fuel received from a fuel source **96** to the mixer body **94**. Referring to FIG. 6, the fuel conduit **92** is connected to and may be cantilevered from the mixer body **94**. The mixer body **94** of FIG. 6 extends axially along a centerline axis **98** of the mixer body **94** from an upstream end **100** of the respective fuel-air mixer **66** and its mixer body **94** to a downstream end **102** of the respective fuel-air mixer **66** and its mixer body **94**. The mixer body **94** projects radially out to an outer side of the mixer body **94**. The mixer body **94** of FIG. 6 includes a mixer center body **104**, a mixer fuel injector body **106** and a mixer shroud **108**. The mixer body **94** of FIG. 6 also includes an inner air swirler **110**, an outer air swirler **112** and a fuel swirler **114**.

[0048] The mixer center body **104** extends axially along the centerline axis **98** from an upstream end **116** of the mixer center body **104** to a downstream end **118** of the mixer center body **104**. The center body upstream end **116** may be axially recessed from the mixer upstream end **100**. The center body downstream end **118** may be axially recessed from the mixer downstream end **102**. The mixer center body **104** may be configured as a sound resonator **120**; e.g., a sound attenuator, a sound muffler, etc. The mixer center body **104** of FIG. 6, for example, includes a center body

upstream endwall **122**, a center body downstream endwall **124** and a tubular center body sidewall **126**.

[0049] The upstream endwall **122** is disposed at the center body upstream end **116**. The upstream endwall **122** projects radially to the center body sidewall **126**. The upstream endwall **122** includes a single air inlet **128** which projects axially through the upstream endwall **122**. The air inlet **128** of FIG. **6** is coaxial with the centerline axis **98**. The air inlet **128** has a cross-sectional area when viewed, for example, in a reference plane perpendicular to the centerline axis **98**. Of course, in other embodiments, it is contemplated the upstream endwall **122** may alternatively include multiple of the air inlets.

[0050] The downstream endwall **124** is disposed at the center body downstream end **118**. The downstream endwall **124** projects radially to the center body sidewall **126**. The downstream endwall **124** includes one or more air outlets **130** which project axially through the downstream endwall **124**. Referring to FIG. **7**, these air outlets **130** may be arranged in one or more concentric arrays or otherwise distributed in, for example, a showerhead pattern. Referring again to FIG. **6**, each of the air outlets **130** has a (e.g., common) cross-sectional area when viewed, for example, in a reference plane perpendicular to the centerline axis **98**. This air outlet cross-sectional area is sized smaller than the air inlet cross-sectional area. However, a total of the air outlet cross-sectional areas (a total air outlet cross-sectional area) may be sized greater than the air inlet cross-sectional area. The total air outlet cross-sectional area, for example, may be two times ($2\times$), three times ($3\times$), four times ($4\times$) or more than the air inlet cross-sectional area. Note, where the mixer center body **104** includes multiple of the air inlets, the total air outlet cross-sectional area may be two times ($2\times$), three times ($3\times$), four times ($4\times$) or more than a total of the air inlet cross-sectional areas (a total air inlet cross-sectional area).

[0051] The center body sidewall **126** extends axially between and is connected to the upstream endwall **122** and the downstream endwall **124**. The center body sidewall **126** of FIG. **6** includes a sidewall upstream section **132** and a sidewall downstream section **134**. The sidewall upstream section **132** projects axially along the centerline axis **98** between the upstream endwall **122** and the sidewall downstream section **134**. This sidewall upstream section **132** may have a cylindrical geometry. A radius from the centerline axis **98** to the center body sidewall **126**, for example, may be uniform (e.g., constant) as the sidewall upstream section **132** extends axially along the centerline axis **98** from the upstream endwall **122** to the sidewall downstream section **134**. The sidewall downstream section **134** projects axially along the centerline axis **98** between the downstream endwall **124** and the sidewall upstream section **132**. This sidewall downstream section **134** may have a frustoconical geometry. The radius from the centerline axis **98** to the center body sidewall **126**, for example, may change (e.g., continuously or incrementally increase) as the sidewall downstream section **134** extends axially along the centerline axis **98** from the sidewall upstream section **132** to the downstream endwall **124**.

[0052] The center body elements **122**, **124** and **126** of FIG. **6** collectively form an internal cavity **136** (e.g., a resonance chamber) within the mixer center body **104**. This center body cavity **136** extends axially within the mixer center body **104** along the centerline axis **98** from the upstream endwall **122** to the downstream endwall **124**. The center body cavity **136** thereby fluidly couples the air inlet **128** to the air outlets **130**. The center body cavity **136** also projects radially within the mixer center body **104** from the centerline axis **98** to the center body sidewall **126**. An upstream section **138** of the center body cavity **136** extends axially along the sidewall upstream section **132** and may have a cylindrical geometry. A downstream section **140** of the center body cavity **136** extends axially along the sidewall downstream section **134** and may have a frustoconical geometry. The cavity downstream section **140** of FIG. **6**, for example, radially expands as the center body cavity **136** extends axially from the cavity upstream section **138** to (or about) the downstream endwall **124** and its air outlets **130**.

[0053] The fuel injector body **106** extends axially along the centerline axis **98** from the mixer

upstream end **100** to a downstream end **142** of the fuel injector body **106**. The injector body downstream end **142** may be axially recessed from the mixer downstream end **102**. The injector body downstream end **142** may also be axially recessed from the center body downstream end **118**. The fuel injector body **106** extends radially from a radial inner side **144** of the fuel injector body **106** to a radial outer side **146** of the fuel injector body **106**. The injector body inner side **144** is spaced radially outward from a radial outer side **148** of the mixer center body **104** and its center body sidewall **126**. The fuel injector body **106** extends circumferentially about (e.g., completely around) the centerline axis **98**. With this arrangement, the fuel injector body **106** axially overlaps and circumscribes the mixer center body **104**.

[0054] The fuel injector body **106** forms a mixer inlet **150** (e.g., an airflow inlet) within the respective fuel-air mixer **66** and its mixer body **94**. This mixer inlet **150** projects axially along the centerline axis **98** into the respective fuel-air mixer **66** and its mixer body **94** from the mixer upstream end **100** to the mixer center body **104** and its upstream endwall **122**. The mixer inlet **150** thereby fluidly couples the diffuser plenum **62** to the air inlet **128**. The mixer inlet **150** also fluidly couples the diffuser plenum **62** to an annular inner passage **152** within the respective fuel-air mixer **66** and its mixer body **94**.

[0055] The inner passage **152** extends axially along the centerline axis **98** within the respective fuel-air mixer **66** and its mixer body **94** from the mixer inlet **150** towards a mixer outlet **154** from the respective fuel-air mixer **66** and its mixer body **94** into the combustion chamber **60**. The inner passage **152** of FIG. **6**, for example, extends axially along the mixer center body **104** and the fuel injector body **106** from the mixer inlet **150** to an annular intermediate passage **156**. A radial inner peripheral boundary of the inner passage **152** is formed by the mixer center body **104** at its center body outer side **148**. A radial outer peripheral boundary of the inner passage **152** is formed by the fuel injector body **106** at its injector body inner side **144**.

[0056] An upstream section **158** of the inner passage **152** extends axially along the sidewall upstream section **132** and may have an annular cylindrical geometry. For example, the inner passage **152** has a cross-sectional area when viewed, for example, in a reference plane perpendicular to the centerline axis **98**. This inner passage cross-sectional area may remain uniform as the inner passage upstream section **158** extends axially from the mixer inlet **150** to a downstream section **160** of the inner passage **152**.

[0057] The inner passage downstream section **160** extends axially along the sidewall downstream section **134** and may have an annular frustoconical geometry. The inner passage cross-sectional area, for example, may change (e.g., decrease) as the inner passage downstream section **160** extends axially from the inner passage upstream section **158** to (or about) the intermediate passage **156**. In addition or alternatively, an inner radius and/or an outer radius of the inner passage downstream section **160** may change (e.g., increase) as the inner passage downstream section **160** extends axially from the inner passage upstream section **158** to (or about) the intermediate passage **156**. The inner passage **152** and, more particularly, its downstream section **160** may thereby radially taper and/or diverge radially outward away from the centerline axis **98** as the inner passage **152** and its downstream section **160** extend axially towards the mixer outlet **154**; e.g., to the intermediate passage **156**. The inner passage upstream section **158** therefore has a cross-sectional area that is greater than (e.g., between 1.05 to 1.20 times greater than) a cross-sectional area of the inner passage downstream section **160** (e.g., adjacent the intermediate passage **156**).

[0058] A centerline **162** of a (e.g., top or bottom radial) half of the inner passage downstream section **160** of FIG. **6** is angularly offset from a centerline **164** of a corresponding (e.g., top or bottom radial) half of the inner passage upstream section **158** by an included obtuse angle **166** when viewed, for example, in a reference plane parallel with (e.g., including) the centerline axis **98**. This obtuse angle **166** is less than one-hundred and eighty degrees ($<180^\circ$) and may be equal to or greater than one-hundred and thirty-five degrees ($\geq 135^\circ$). The downstream section centerline **162** of FIG. **6** is also angularly offset from the centerline axis **98** by an included (e.g., non-zero) acute

angle **168**. This acute angle **168** is greater than zero degrees ($>0^\circ$) and may be equal to or less than forty-five degrees ($\leq 45^\circ$). The upstream section centerline **164** of FIG. **6** is parallel with the centerline axis **98**; however, the present disclosure is not limited thereto.

[0059] The fuel injector body **106** includes an internal fuel plenum **170** and an internal annular fuel passage **172**. The fuel plenum **170** fluidly couples the fuel conduit **92** to the fuel passage **172**. The fuel plenum **170** may axially overlap the mixer inlet **150** and/or the inner passage upstream section **158**. The fuel plenum **170** may also circumscribe the mixer inlet **150** and/or the inner passage upstream section **158**. The fuel passage **172** extends axially within the fuel injector body **106** from the fuel plenum **170** to an annular outlet **174** from the fuel passage **172** into the inner passage **152**. This fuel passage outlet **174** is formed in a canted (e.g., frustoconical) surface of the fuel injector body **106** along the injector body inner side **144**. The fuel passage outlet **174** of FIG. **6** is disposed axially at (e.g., on, adjacent or proximate) the injector body downstream end **142**.

[0060] The fuel passage **172** of FIG. **6** includes an upstream section **176**, a downstream section **178** and an intermediate section **180**. The fuel passage upstream section **176** extends axially from the fuel plenum **170** to the fuel passage intermediate section **180**. This fuel passage upstream section **176** may have a uniform cross-sectional area along its longitudinal length; e.g., axial length. The fuel passage downstream section **178** extends axially from the fuel passage intermediate section **180** to the fuel passage outlet **174**. This fuel passage downstream section **178** may have a uniform cross-sectional area along its longitudinal length; e.g., axial length. The fuel passage intermediate section **180** extends axially between and to the fuel passage upstream section **176** and the fuel passage downstream section **178**. This fuel passage intermediate section **180** may radially taper inwards such that the cross-sectional area of the fuel passage downstream section **178** is less than the cross-sectional area of the fuel passage upstream section **176**. However, in other embodiments, it is contemplated the fuel passage **172** may alternatively have a uniform cross-sectional area from the fuel plenum **170** to the fuel passage outlet **174**.

[0061] A centerline **182** of a (e.g., top or bottom radial) half of the fuel passage **172** of FIG. **6** and its downstream section **178** is arranged parallel with the centerline axis **98** when viewed, for example, in a reference plane parallel with (e.g., including) the centerline axis **98**; however, the present disclosure is not limited thereto. The fuel passage centerline **182** of FIG. **6** is also angularly offset from the downstream section centerline **162** by an included (e.g., non-zero) acute angle **184**. This acute angle **184** is greater than zero degrees ($>0^\circ$) and may be equal to or less than thirty degrees ($\leq 30^\circ$).

[0062] The mixer shroud **108** extends axially along the centerline axis **98** from an upstream end **186** of the mixer shroud **108** to the mixer downstream end **102**. The shroud upstream end **186** is axially spaced from the mixer upstream end **100**. The mixer shroud **108** extends radially from a radial inner side **188** of the mixer shroud **108** to a radial outer side **190** of the mixer shroud **108**. The shroud inner side **188** is spaced radially outward from injector body outer side **146**. The mixer shroud **108** extends circumferentially about (e.g., completely around) the centerline axis **98**. With this arrangement, the mixer shroud **108** axially overlaps and circumscribes the fuel injector body **106** as well as the mixer center body **104**.

[0063] The mixer shroud **108** forms the mixer outlet **154** within the respective fuel-air mixer **66** and its mixer body **94**. This mixer outlet **154** projects axially along the centerline axis **98** into the respective fuel-air mixer **66** and its mixer body **94** from the mixer downstream end **102** to the mixer center body **104** and its downstream endwall **124**. The mixer outlet **154** thereby fluidly couples the combustion chamber **60** to the air outlets **130**. The mixer outlet **154** also fluidly couples the combustion chamber **60** to the intermediate passage **156**.

[0064] The intermediate passage **156** extends axially along the centerline axis **98** within the respective fuel-air mixer **66** and its mixer body **94** from the mixer outlet **154** to the inner passage **152**. The intermediate passage **156** also extends axially along the centerline axis **98** within the respective fuel-air mixer **66** and its mixer body **94** from the mixer outlet **154** to an annular outer

passage **192** within the respective fuel-air mixer **66** and its mixer body **94**. With the arrangement of FIG. **6**, the inner passage **152** and the outer passage **192** are fluidly coupled to the intermediate passage **156** in parallel. A radial inner peripheral boundary of the intermediate passage **156** is formed by the mixer center body **104** at its center body outer side **148**. A radial outer peripheral boundary of the intermediate passage **156** is formed by the mixer shroud **108** at its shroud inner side **188**. Of course, in other embodiments, the intermediate passage **156** may be omitted and the inner passage **152** and the outer passage **192** may be fluidly coupled to the mixer outlet **154** in parallel.

[0065] The outer passage **192** extends axially along the centerline axis **98** within the respective fuel-air mixer **66** and its mixer body **94** from an annular inlet orifice **194** towards the mixer outlet **154**. The outer passage **192** of FIG. **6**, for example, extends axially along the fuel injector body **106** and the mixer shroud **108** from the inlet orifice **194** to the intermediate passage **156**. A radial inner peripheral boundary of the outer passage **192** is formed by the fuel injector body **106** at its injector body outer side **146**. A radial outer peripheral boundary of the outer passage **192** is formed by the mixer shroud **108** at its shroud inner side **188**. With this arrangement, the fuel injector body **106** may form an annular splitter for directing the compressed core air in the diffuser plenum **62** into the inner passage **152** and the outer passage **192**.

[0066] The outer passage **192** may have an annular frustoconical geometry. For example, the outer passage **192** has a cross-sectional area when viewed, for example, in a reference plane perpendicular to the centerline axis **98**. This outer passage cross-sectional area may change (e.g., decrease) as the outer passage **192** extends axially from (or about) the inlet orifice **194** to (or about) the intermediate passage **156**. In addition or alternatively, an inner radius and/or an outer radius of the outer passage **192** may change (e.g., decrease) as the outer passage **192** extends axially from (or about) the inlet orifice **194** to (or about) the intermediate passage **156**. The outer passage **192** may thereby converge radially inward towards the centerline axis **98** as the outer passage **192** extends axially towards the mixer outlet **154**; e.g., to the intermediate passage **156**.

[0067] A centerline **196** of a (e.g., top or bottom radial) half of the outer passage **192** of FIG. **6** is angularly offset from the centerline axis **98** by an included (e.g., non-zero) acute angle. This acute angle is greater than zero degrees ($>0^\circ$) and may be equal to or less than forty-five degrees ($\leq 45^\circ$). The outer passage centerline **196** of FIG. **6** is also angularly offset from the downstream section centerline **162** by an include (e.g., non-zero) acute angle **198**. This acute angle **198** is greater than zero degrees ($>0^\circ$) and may be equal to or less than forty-five degrees ($\leq 45^\circ$); e.g., between ten degrees (10°) and thirty degrees (30°).

[0068] The inner air swirler **110** may be configured as an axial air swirler. The inner air swirler **110** of FIG. **6**, for example, includes a plurality of inner swirler vanes **200** arranged circumferentially about the centerline axis **98** in an inner vane array. This inner vane array and its inner swirler vanes **200** are arranged within the inner passage **152** and, more particularly, the inner passage upstream section **158**. Each of the inner swirler vanes **200** projects radially across the inner passage **152** and its inner passage upstream section **158** from the mixer center body **104** to the fuel injector body **106**. The inner air swirler **110** and its inner swirler vanes **200** are configured to swirl fluid (e.g., the compressed core air) flowing thereacross in a first circumferential direction (e.g., clockwise or counterclockwise) about the centerline axis **98**.

[0069] The outer air swirler **112** may be configured as an axial air swirler. The outer air swirler **112** of FIG. **6**, for example, includes a plurality of outer swirler vanes **202** arranged circumferentially about the centerline axis **98** in an outer vane array. This outer vane array and its outer swirler vanes **202** are arranged within the outer passage **192**, for example at or near the inlet orifice **194**. Each of the outer swirler vanes **202** projects radially across the outer passage **192** from the fuel injector body **106** to the mixer shroud **108**. The outer air swirler **112** and its outer swirler vanes **202** are configured to swirl fluid (e.g., the compressed core air) flowing thereacross in the first circumferential direction (e.g., clockwise or counterclockwise) about the centerline axis **98**, or

alternatively in a second circumferential direction (e.g., counterclockwise or clockwise) about the centerline axis **98** opposite of the first circumferential direction.

[0070] However, in general, a degree of swirl induced by the outer air swirler **112** is less than a degree of swirl induced by the inner air swirler **110**. To this point, it is contemplated the outer swirler vanes **202** may be replaced with a plurality of non-swirling struts; e.g., struts with mean lines which run longitudinally parallel with the centerline axis **98**. Thus, the outer air swirler **112** may be replaced by a non-swirling vane array.

[0071] The fuel swirler **114** may be configured as an axial fuel swirler. The fuel swirler **114** of FIG. **6**, for example, includes a plurality of fuel swirler vanes **204** arranged circumferentially about the centerline axis **98** in a fuel swirler vane array. This fuel swirler vane array and its fuel swirler vanes **204** are arranged within the fuel passage **172** and, more particularly, the fuel passage upstream section **176**. Each of the fuel swirler vanes **204** projects radially across the fuel passage **172** and its fuel passage upstream section **176**. The fuel swirler **114** and its fuel swirler vanes **204** are configured to swirl fluid (e.g., the fuel) flowing thereacross in the first circumferential direction (e.g., clockwise or counterclockwise) about the centerline axis **98**. In general, a degree of swirl induced by the fuel swirler **114** is similar to the degree of swirl induced by the inner air swirler **110**.

[0072] Referring to FIG. **2**, each fuel-air mixer **66** is mated with the combustor **64**. More particularly, each fuel-air mixer **66** and its mixer body **94** are mated with the bulkhead **68**. The mixer body **94** of FIG. **2**, for example, projects axially along its centerline axis **98** through (or partially into) an aperture in the bulkhead **68**. Each fuel-air mixer **66** and its mixer body **94** may be attached to the combustor **64** and its bulkhead **68** using various techniques; e.g., bonding, mechanical fastening, etc. Each fuel-air mixer **66** and its mixer body **94** may thereby be fixedly attached to the bulkhead **68**. However, in other embodiments, each fuel-air mixer **66** may be moveably attached to the bulkhead **68** through, for example, a sliding guide plate.

[0073] During operation of the combustor section **30** of FIG. **2**, each fuel-air mixer **66** receives the fuel from the fuel source **96** and the compressed core air from the diffuser plenum **62**. At each fuel-air mixer **66** of FIG. **6**, the fuel injector body **106** injects a swirled annular fuel flow out of the fuel passage **172** and into the inner passage **152**. This annular fuel flow is subsequently directed through the intermediate passage **156** and the mixer outlet **154** into the combustion chamber **60**. A radial inner periphery of the annular fuel flow is shrouded by a swirled annular inner airflow, and a radial outer periphery of the annular fuel flow is shrouded by a swirled (or non-swirled) annular outer airflow. These annular airflows flow along the annular fuel flow through the intermediate passage **156** and the mixer outlet **154** into the combustion chamber **60** which may facilitate deeper penetration of the fuel into the combustion chamber **60** before mixing with the compressed air and igniting. However, the swirling of the fuel and the air from the inner passage **152** may also facilitate mixing of the fuel and the air after penetration into the combustion chamber **60**.

[0074] The combustion of the fuel-air mixture within the combustion chamber **60** generates noise. At least some frequencies of this combustion noise (e.g., frequencies equal to or above 700 Hz or 1000 Hz) may be attenuated by each sound resonator **120**. For example, noise waves generated by the combustion process may travel into the center body cavity **136** through the air outlets **130**. These noise waves may be captured and muffled within the respective sound resonator **120**. Note, an acoustic impedance and/or damping effectiveness of each sound resonator **120** may be tuned by selectively tailoring a size (e.g., diameter) of the air outlets **130** and/or a quantity of the air outlets **130**; e.g., a porosity of the respective downstream endwall **124**.

[0075] The fuel source **96** of FIG. **2** includes a fuel reservoir **206** and/or a fuel flow regulator **208**; e.g., a valve and/or a pump. The fuel reservoir **206** is configured to store the fuel before, during and/or after turbine engine operation. The fuel reservoir **206**, for example, may be configured as or otherwise include a tank, a cylinder, a pressure vessel, a bladder or any other type of fuel storage container. The fuel flow regulator **208** is configured to direct and/or meter a flow of the fuel from the fuel reservoir **206** to one or more or all of the fuel-air mixers **66**.

[0076] The fuel delivered by the fuel source **96** may be a non-hydrocarbon fuel; e.g., a hydrocarbon free fuel. An example of the non-hydrocarbon fuel is hydrogen fuel; e.g., hydrogen (H.sub.2) gas. The turbine engine **20** of FIG. **1** may thereby be configured as a non-hydrocarbon turbine engine; e.g., a hydrocarbon free turbine engine. The present disclosure, however, is not limited to non-hydrocarbon turbine engines. The fuel delivered by the fuel source **96**, for example, may alternatively be a hydrocarbon fuel such as, but not limited to, kerosene or jet fuel. The turbine engine **20** of FIG. **1** may thereby be configured as a hydrocarbon turbine engine. Alternatively, the fuel source **96** may be configured as a multi-fuel system operable to deliver, individually or in combination, multiple different fuels (e.g., a non-hydrocarbon fuel and a hydrocarbon fuel, etc.) for combustion within the combustion chamber **60**. The turbine engine **20** of FIG. **1** may thereby be configured as a multi-fuel turbine engine; e.g., a dual-fuel turbine engine. However, for ease of description, the fuel delivered by the fuel source **96** may be described as the non-hydrocarbon fuel; e.g., the hydrogen fuel.

[0077] In some embodiments, referring to FIG. **6**, each fuel-air mixer **66** or at least its mixer body **94** may be formed as a monolithic body. The term “monolithic” may describe a body which is cast, machined, additively manufactured and/or otherwise formed as a single, integral unit. By contrast, a non-monolithic body includes multiple bodies which are separately formed and then mechanically fastened and/or otherwise attached to one another after the formation of those bodies. Of course, in other embodiments, it is contemplated one or more elements (e.g., **104**, **106**, **108**, **110**, **112** and/or **114**) of each fuel-air mixer **66** or its mixer body **94** may be discretely formed and then attached to one another. The present disclosure therefore is not limited to such an exemplary monolithic configuration.

[0078] In some embodiments, referring to FIGS. **8A** and **8B**, the inner fuel-air mixers **66A** and the outer fuel-air mixers **66B** may be arranged into one or more clusters **210**; e.g., local arrays and/or matrices. Each cluster **210** of FIGS. **8A** and **8B**, for example, includes a respective pair of the inner fuel-air mixers **66A** and a (e.g., corresponding circumferentially aligned) respective pair of the outer fuel-air mixers **66B**. One or more of these clusters **210** may also be associated with a respective pilot fuel injector **212**. The fuel-air mixers **66** in each cluster of FIGS. **8A** and **8B**, for example, are arranged in an array about the respective pilot fuel injector **212**. In some embodiments, referring to FIG. **8A**, the array of the fuel-air mixers **66** in the respective cluster **210** may be symmetrical; e.g., a symmetric array. In other embodiments, referring to FIG. **8B**, the array of the fuel-air mixers **66** in the respective cluster **210** may be asymmetrical; e.g., an asymmetrical array.

[0079] In some embodiments, referring to FIG. **6**, a downstream, radial inner corner of the mixer shroud **108** may be eased; e.g., chamfered. A downstream, radial outer corner of the fuel injector body **106** may also or alternatively be eased; e.g., chamfered. In other embodiments, referring to FIG. **9**, the downstream, radial inner corner of the mixer shroud **108** may be sharp; e.g., pointed, not-chamfered, etc. The downstream, radial outer corner of the fuel injector body **106** may also or alternatively be sharp; e.g., pointed, not-chamfered, etc.

[0080] The fuel-air mixer(s) **66** may be included in various turbine engines other than the one described above. The fuel-air mixer(s) **66**, for example, may be included in a geared turbine engine where a geartrain connects one or more shafts to one or more rotors in a fan section, a compressor section and/or any other engine section. Alternatively, the fuel-air mixer(s) **66** may be included in a turbine engine configured without a geartrain; e.g., a direct drive turbine engine. The fuel-air mixer(s) **66** may be included in a geared or non-geared turbine engine configured with a single spool, with two spools (e.g., see FIG. **1**), or with more than two spools. The turbine engine may be configured as a turbofan engine, a turbojet engine, a turboprop engine, a turboshaft engine, a propfan engine, a pusher fan engine or any other type of turbine engine. The turbine engine may alternatively be configured as an auxiliary power unit (APU) or an industrial gas turbine engine. The present disclosure therefore is not limited to any particular types or configurations of turbine

engines.

[0081] While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. An apparatus for a turbine engine, comprising: a fuel-air mixer including a mixer outlet, an annular inner passage, an air swirler, an annular fuel passage, a center body and a fuel swirler; the annular inner passage extending axially along an axis within the fuel-air mixer, the annular inner passage extending axially along and circumscribing the center body, the annular inner passage including an inner passage downstream section and an inner passage upstream section fluidly coupled with the mixer outlet through the inner passage downstream section, and the inner passage downstream section radially tapering and diverging radially outward away from the axis as the annular inner passage extends axially towards the mixer outlet; the air swirler disposed with the inner passage upstream section; the annular fuel passage circumscribing the annular inner passage and extending axially within the fuel-air mixer to the inner passage downstream section, wherein a centerline of a half of the annular fuel passage, at an annular outlet from the annular fuel passage into the inner passage downstream section, is parallel with the axis; the center body including an air inlet, a plurality of air outlets and an internal cavity, the internal cavity fluidly coupled with and between the air inlet and the plurality of air outlets, the internal cavity fluidly coupled with the mixer outlet through the plurality of air outlets, and the internal cavity extending within the axis; and the fuel swirler disposed with the annular fuel passage.
2. The apparatus of claim 1, wherein a cross-sectional area of the inner passage downstream section decreases as the annular inner passage extends axially towards the mixer outlet.
3. The apparatus of claim 1, wherein a centerline of a half of the inner passage upstream section is parallel with the axis; and a centerline of a half of the inner passage downstream section is angularly offset from the axis by an acute angle.
4. The apparatus of claim 1, wherein the centerline of the half of the annular fuel passage is angularly offset from a centerline of a half of the inner passage downstream section by an acute angle.
5. (canceled)
6. The apparatus of claim 1, wherein the air swirler is configured to swirl air flowing within the annular inner passage towards the mixer outlet in a first circumferential direction about the axis; and the fuel swirler is configured to swirl fuel flowing within the annular fuel passage towards the inner passage downstream section in the first circumferential direction about the axis.
7. The apparatus of claim 1, wherein the fuel-air mixer further includes an annular outer passage circumscribing the annular inner passage and the annular fuel passage; and the annular outer passage extends axially along the axis within the fuel-air mixer and is fluidly coupled with the mixer outlet.
8. The apparatus of claim 7, wherein the annular outer passage converges radially inward towards the axis as the annular outer passage extends axially towards the mixer outlet.
9. The apparatus of claim 7, wherein the fuel-air mixer further includes a plurality of non-swirling struts arranged circumferentially about the axis and extending radially across the annular outer passage.

10. The apparatus of claim 7, wherein the air swirler is an inner air swirler, and the fuel-air mixer further includes an outer air swirler disposed with the annular outer passage.
11. (canceled)
12. The apparatus of claim 1, wherein the center body forms an inner peripheral boundary of the annular inner passage.
13. The apparatus of claim 1, wherein a cross-sectional area of the air inlet is greater than a cross-sectional area of each of the plurality of air outlets; and the cross-sectional area of the air inlet is less than a total cross-sectional area of the plurality of air outlets.
14. The apparatus of claim 1, wherein the internal cavity includes a cavity upstream section and a cavity downstream section between the cavity upstream section and the plurality of air outlets; and the cavity downstream section radially expands as the internal cavity extends axially within the center body towards the plurality of air outlets.
15. The apparatus of claim 1, further comprising: an annular combustor bulkhead extending circumferentially around an axial centerline; the fuel-air mixer being one of a plurality of fuel-air mixers mounted to the annular combustor bulkhead, and a first of the plurality of fuel-air mixers located radially outboard of a second of the plurality of fuel-air mixers.
16. The apparatus of claim 15, wherein the first of the plurality of fuel-air mixers is circumferentially aligned with the second of the plurality of fuel-air mixers.
17. The apparatus of claim 1, further comprising: a pilot fuel injector; the fuel-air mixer being one of a plurality of fuel-air mixers arranged in an array symmetrically about the pilot fuel injector.
18. The apparatus of claim 1, further comprising: a pilot fuel injector; the fuel-air mixer being one of a plurality of fuel-air mixers arranged in an array asymmetrically about the pilot fuel injector.
19. An apparatus for a turbine engine, comprising: a fuel-air mixer including an annular intermediate passage, an annular inner passage, an annular outer passage and an annular fuel passage; the annular inner passage extending axially along an axis within the fuel-air mixer to the annular intermediate passage, the annular inner passage diverging radially outward away from the axis as the annular inner passage extends axially to the annular intermediate passage; the annular outer passage extending axially along the axis within the fuel-air mixer to the annular intermediate passage, the annular outer passage circumscribing the annular inner passage and the annular fuel passage, and the annular outer passage converging radially inward towards the axis as the annular outer passage extends axially from an annular inlet orifice to the annular intermediate passage; and the annular fuel passage radially between the annular inner passage and the annular outer passage, the annular fuel passage circumscribing the annular inner passage, the annular fuel passage extending axially along the axis within the fuel-air mixer to an annular outlet from the annular fuel passage into the annular inner passage, and a centerline of the annular outlet arranged parallel with the axis along an axial length of the annular outlet.
20. An apparatus for a turbine engine, comprising: a fuel-air mixer including a mixer outlet, an annular inner passage, an annular fuel passage and a resonator; the annular inner passage extending axially along an axis within the fuel-air mixer and circumscribing the resonator, the annular inner passage including an inner passage downstream section and an inner passage upstream section fluidly coupled with the mixer outlet through the inner passage downstream section, and the inner passage downstream section diverging radially outward away from the axis as the annular inner passage extends axially towards the mixer outlet; the annular fuel passage circumscribing the annular inner passage and extending axially within the fuel-air mixer to the inner passage downstream section; and the resonator forming an inner peripheral boundary of the annular inner passage, the resonator including an air inlet, a plurality of air outlets and an internal cavity fluidly coupled with and between the air inlet and the plurality of air outlets, the internal cavity fluidly coupled with the mixer outlet through the plurality of air outlets, and the internal cavity extending within the axis.
21. The apparatus of claim 1, wherein the annular fuel passage includes an upstream section, a

downstream section and an intermediate section; the upstream section includes a first cross-sectional area; the downstream section includes a second cross-sectional area that is different than the first cross-sectional area; and the first cross-sectional area tapers to the second cross-sectional area at the intermediate section.
